

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Applied Business

Practical – IX

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9. Hypothesis Testing:

a. Formulate null and alternative hypotheses related to a business problem, conduct a hypothesis test using appropriate statistical tests, and interpret the Results.

Dataset: supermarket_sales.csv

1. Load the supermarket_sales.csv dataset.
2. Display the first five records.

```
import pandas as pd
from scipy.stats import ttest_1samp

# 1. Load dataset
df = pd.read_csv("supermarket_sales.csv")

# 2. Display first five records
print("First Five Records:")
print(df.head())
```

First Five Records:

	invoice_id	revenue
0	1001	1200
1	1002	1450
2	1003	1600
3	1004	1800
4	1005	1350

3. Select the revenue column.
4. Check for missing values.
5. Calculate the mean revenue.
6. Assume that the expected average revenue is ₹1,500.

```
# 3. Select revenue column
revenue = df["revenue"]

# 4. Check for missing values
print("\nMissing Values:")
print(revenue.isnull().sum())

# 5. Calculate mean revenue
mean_revenue = revenue.mean()
print("\nMean Revenue:", mean_revenue)

# 6. Expected average revenue
expected_mean = 1500
```

Mean Revenue: 1541.6666666666667

7. Write the null hypothesis:

$H_0: \mu = 1500$

8. Write the alternative hypothesis:

$H_1: \mu \neq 1500$

9. Set the significance level as 5%.

10. Perform a one-sample t-test.

```
# 7 & 8. Hypotheses
print("\nH0:  $\mu = 1500$ ")
print("\nH1:  $\mu \neq 1500$ ")

# 9. Significance level
alpha = 0.05

# 10. One-sample t-test
t_stat, p_value = ttest_1samp(revenue, expected_mean)
print("\nT-Statistic:", t_stat)

# 11. P-value
print("P-Value:", p_value)
```

T-Statistic: 1.2819978055684943
P-Value: 0.2390881851286923

11. Calculate the p-value.

12. Compare the p-value with 0.05.

13. State whether the null hypothesis is rejected or not rejected.

14. Write the conclusion in simple business terms.

```
# 12 & 13. Compare p-value
if p_value < alpha:
    print("\nResult: Reject the Null Hypothesis")
else:
    print("\nResult: Do Not Reject the Null Hypothesis")

# 14. Conclusion
if p_value < alpha:
    print("Conclusion: The average revenue is significantly different from ₹1,500.")
else:
    print("Conclusion: There is not enough evidence to say that the average revenue is different from ₹1,500.")
```

Result: Do Not Reject the Null Hypothesis

Conclusion: There is not enough evidence to say that the average revenue is different from ₹1,500.

b. Perform A/B testing on a website or marketing campaign to evaluate the effectiveness of different strategies and make data-driven decisions

Dataset: ab_testing.csv

1. Load the A/B testing dataset.

2. Display the data.

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Load the dataset
data = pd.DataFrame({
    'Group': ['A', 'B'],
    'Users': [1000, 1000],
    'Conversions': [120, 150]
})
```

```
# 2. Display the data
print(data)
```

	Group	Users	Conversions
0	A	1000	120
1	B	1000	150

3. Calculate the conversion rate for Group A.

```
# 3. Conversion rate for Group A
conv_rate_A = data.loc[data['Group'] == 'A', 'Conversions'].values[0] / data.loc[data['Group'] == 'A', 'Users'].values[0]
print("Conversion Rate A:", conv_rate_A)
```

✓ 0.0s

Conversion Rate A: 0.12

4. Calculate the conversion rate for Group B.

```
# 4. Conversion rate for Group B
conv_rate_B = data.loc[data['Group'] == 'B', 'Conversions'].values[0] / data.loc[data['Group'] == 'B', 'Users'].values[0]
print("Conversion Rate B:", conv_rate_B)
```

✓ 0.0s

Conversion Rate B: 0.15

5. Compare the conversion rates of A and B

```
# 5. Compare conversion rates
print("Difference:", conv_rate_B - conv_rate_A)
```

✓ 0.0s

Difference: 0.03

6. Write the null hypothesis:

$H_0: p_A = p_B$

```
# 6. Null Hypothesis
print("H0: pA = pB")
```

✓ 0.0s

H0: pA = pB

7. Write the alternative hypothesis:

$H_1: p_A \neq p_B$

```
# 7. Alternative Hypothesis
print("H1: pA ≠ pB")
```

✓ 0.0s

H1: pA ≠ pB

8. Set the significance level as 5%.

```
# 8. Significance level
alpha = 0.05
print("Significance Level:", alpha)
```

✓ 0.0s

Significance Level: 0.05

9. Perform a two-proportion test.

```
# 9. Perform two-proportion z-test
successes = [data.loc[data['Group'] == 'A', 'Conversions'].values[0],
             data.loc[data['Group'] == 'B', 'Conversions'].values[0]]
trials = [data.loc[data['Group'] == 'A', 'Users'].values[0],
          data.loc[data['Group'] == 'B', 'Users'].values[0]]
```

10. Calculate the p-value

```
# 10. Calculate the p-value
print("p-value:", p_value)



p-value: 0.04964038686536878


```

11. Compare the p-value with 0.05.

```
# 11. Compare the p-value with
alpha = 0.05
if p_value < alpha:
    print("Reject H0")
else:
    print("Do not reject H0")



Reject H0


```

12. State whether the null hypothesis is rejected or not rejected.

```
# 12. State whether the null hypothesis is rejected or not rejected
decision = "Rejected" if p_value < alpha else "Not Rejected"
print("Null Hypothesis:", decision)



Null Hypothesis: Rejected


```

13. Create a bar chart to compare the conversion rates.

```
# 13. Bar chart comparing conversion rates
plt.bar(['Group A', 'Group B'], [conv_rate_A, conv_rate_B], color=['blue', 'green'])
plt.ylabel("Conversion Rate")
plt.title("Conversion Rate Comparison")
plt.show()
```

